

A Semi-Independent Policies Training Method with Shared Representation for Heterogeneous Multi-Agent Reinforcement Learning

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The Problem: Sharing Across Different Agents

- Cooperative multi-agent RL suits traffic control, sensor networks and robot swarms, but training is costly and unstable
- Parameter sharing (one policy trained on every agent's experience) works well when all agents are identical
- Heterogeneous agents differ in inputs, outputs and roles, so one shared network no longer fits

Full parameter sharing

One network for all agents
Fast, but forces agents to behave alike
and needs padded inputs

Independent policies

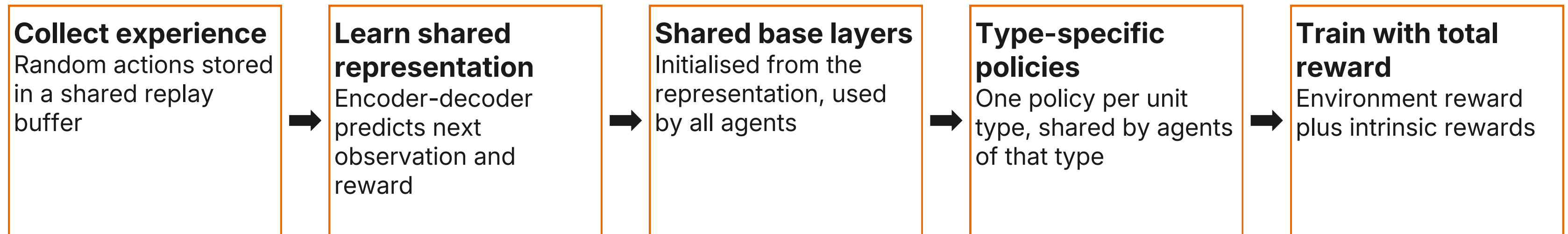
One network per agent
Specialised, but slow to converge and
wastes shared experience

STSR (this paper)

Share fully within a type, partly across
types
Keeps sharing fast while letting each
type specialise

Setting: cooperative Dec-POMDP, where each agent sees only its own partial observation. Example: in the StarCraft II map MMM2, Marines and Marauders attack while the Medivac heals allies.

The STSR Framework



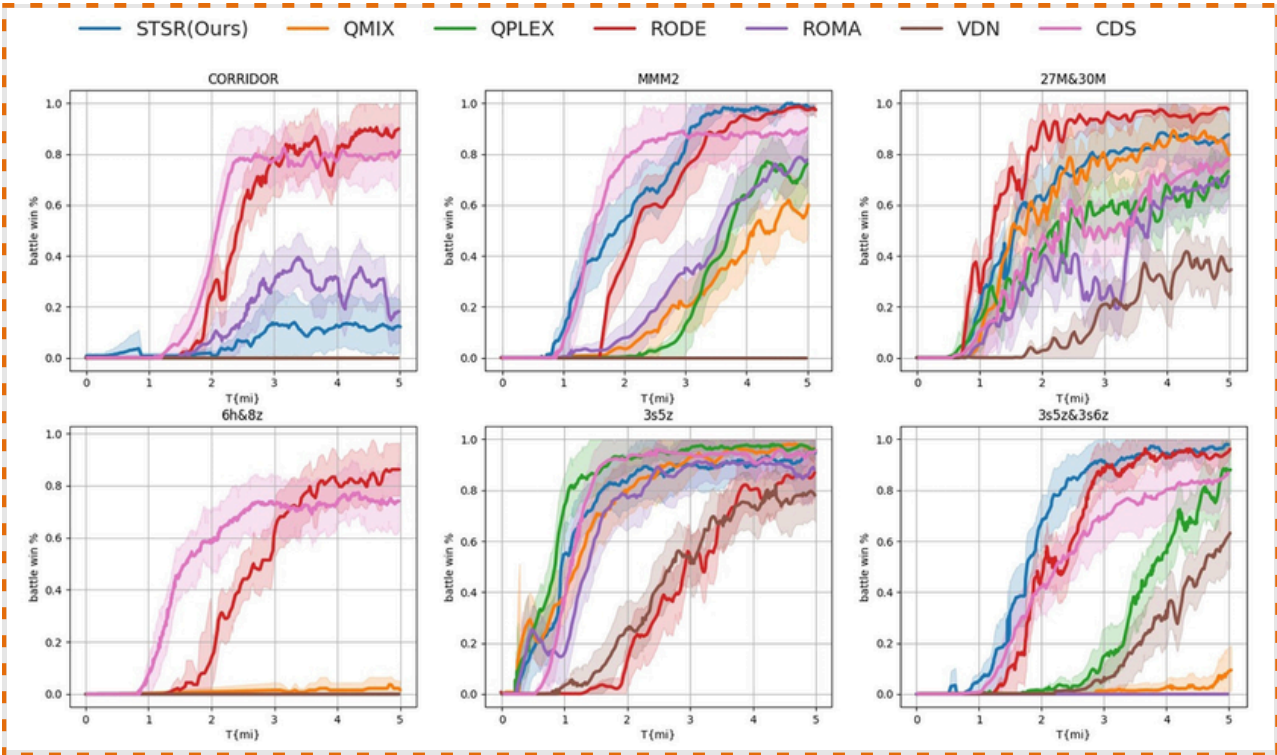
Three components

- **Shared representation:** gives every agent type the same input/output format
- **Semi-independent policies:** full sharing within a type, hard parameter sharing across types
- **Intrinsic rewards:** an exploration bonus from prediction error and a diversity bonus that keeps types distinguishable

Key ideas in equations

- Hard parameter sharing:
 $y1 = g1(f(x)), y2 = g2(f(x))$
f = shared layers, g1 and g2 = task-specific heads (from multi-task learning)
- Total reward:
 $r_{total} = r + \lambda_e \cdot r_{explore} + \lambda_d \cdot r_{diversity}$
Inspired by how the brain reuses abstract knowledge for new tasks

Results



Source: Zhao et al., 2023, Fig. 6.

Experimental setup

Benchmark	StarCraft II multi-agent challenge (SMAC), 14 maps
Map types	Easy, hard and super-hard; homogeneous and heterogeneous teams
Baselines	VDN, QMIX, QPLEX, ROMA, RODE, CDS
Runs	4 random seeds, median reported

- **Best on heterogeneous super-hard maps:** STSR performs best on 3s5z_vs_3s6z and MMM2
- **Faster convergence:** on heterogeneous maps it converges faster than role-based methods
- **Easy maps need less:** QPLEX performs best on the easy map 3s5z, where little exploration is needed
- **Weaker on identical agents:** when all agents are the same type, RODE, ROMA and CDS perform better

Key Insight: Shared Knowledge, Specialised Roles

Types form clear clusters

In MMM2, the learned representations of Marines, Marauders and the Medivac form separate clusters (Fig. 8). The self-supervised embedding is tighter; the RL embedding spreads more to fit individual agents.

Ablation shows what matters

Every component helps. Not updating the shared representation during RL performs worst. The diversity reward and hard sharing help only on heterogeneous maps, and hard sharing mainly speeds up training.

Transfer without retraining

New agents copy their type's policy and representation, so a policy trained on 3s5z_vs_3s6z still performs well on unseen maps with more units (Fig. 9).

Grouping agents by type lets them share experience where they are similar and specialise where they differ.

Takeaways and Future Work

Contributions

- Brings multi-task hard parameter sharing into multi-agent RL
- Supervised shared representation that handles inputs and outputs of any agent type
- Intrinsic rewards built from the representation's prediction error

Limitations

- Agents are grouped only by predefined unit type
- Weaker when all agents are the same type
- Large groups of one type (e.g. 20 Zerglings in bane_vs_bane) may need extra clustering
- Evaluated on one benchmark (SMAC)

Future Work (from the paper)

- Explain why randomly clustering identical agents beat existing methods
- Study the best group size and initial settings for parameter sharing
- Apply the method to curriculum learning, which the authors suggest it supports

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Link to Presentation: <https://canva.link/stsr-presentation-23k-0074>